

CUSTOMER E-COMMERCE SALES ANALYTICS

Business Insights & Strategic Recommendations Report

Prepared By: Himanshu Rawat

Role: Data Analyst Project

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, SciPy

Dataset: Brazilian Olist E-Commerce Dataset

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1. EXECUTIVE SUMMARY

This report analyzes customer transactions, product sales, delivery performance, and customer satisfaction using the Brazilian Olist E-Commerce dataset.

The objective is to identify key business drivers affecting revenue generation, operational efficiency, and customer experience.

The analysis reveals three major findings:

- Revenue is concentrated within a limited number of regions and product categories.
- Delivery performance significantly impacts customer satisfaction.
- Statistical testing confirms that delayed deliveries are associated with lower review scores.

Based on these findings, several strategic recommendations are proposed to improve customer retention, operational performance, and overall profitability.

2. BUSINESS PROBLEM

E-commerce organizations generate large volumes of transactional data but often struggle to convert that data into actionable business insights.

Management seeks to understand:

- Customer purchasing patterns
- Sales performance drivers
- Delivery efficiency
- Customer satisfaction factors

This analysis addresses these challenges through data-driven investigation.

3. DATA OVERVIEW

Dataset Source

Brazilian Olist E-Commerce Dataset

Data Scope

Dataset	Description
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Customers	Customer information
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Orders	Transaction records
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Products	Product details
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Payments	Payment information
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Reviews	Customer reviews
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Sellers	Seller information
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Analysis Period

- **Time Frame:** September 2016 – October 2018
- **Duration :** 25 Months
- **Analysis Level:** Yearly And Quarterly
- **Focus Areas:**
 - Customer Purchasing Behaviour
 - Order Performance
 - Payment Trends
 - Product Demand
 - Customer Satisfaction

4. METHODOLOGY

The project followed the following analytical workflow:

Phase 1: Data Preparation

- Missing value treatment
- Data type correction
- Duplicate removal
- Data quality validation

Phase 2: Exploratory Analysis

- Customer analysis
- Sales analysis
- Logistics analysis
- Customer satisfaction analysis

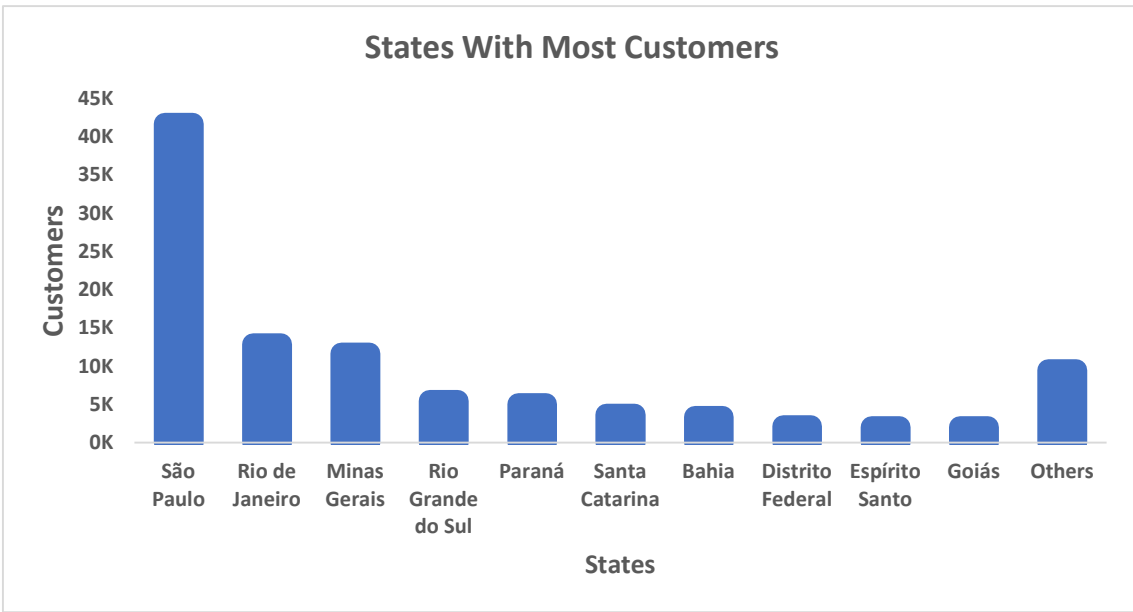
Phase 3: Statistical Testing

- Welch's T-Test
 - Z-Test
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5. CUSTOMER ANALYTICS

Regional Customer Distribution

Top 10 State with most Customers



Findings

- São Paulo represents the largest customer base consists of 42% of total customers.
- Out of 28 regions top 10 region contributes more than 70% of customer base.
- Customer concentration is highly skewed toward a few regions.
- Several states with less customers are Pará, Mato Grosso, Maranhão, Mato Grosso do Sul, Paraíba, Piauí, Rio Grande do Norte, Alagoas, Sergipe, Tocantins, Rondônia, Amazonas and Acre.

Business Implications

Marketing resources should prioritize high-performing regions while identifying growth opportunities in underperforming markets.

6. SALES ANALYTICS

Product Wise Order Share



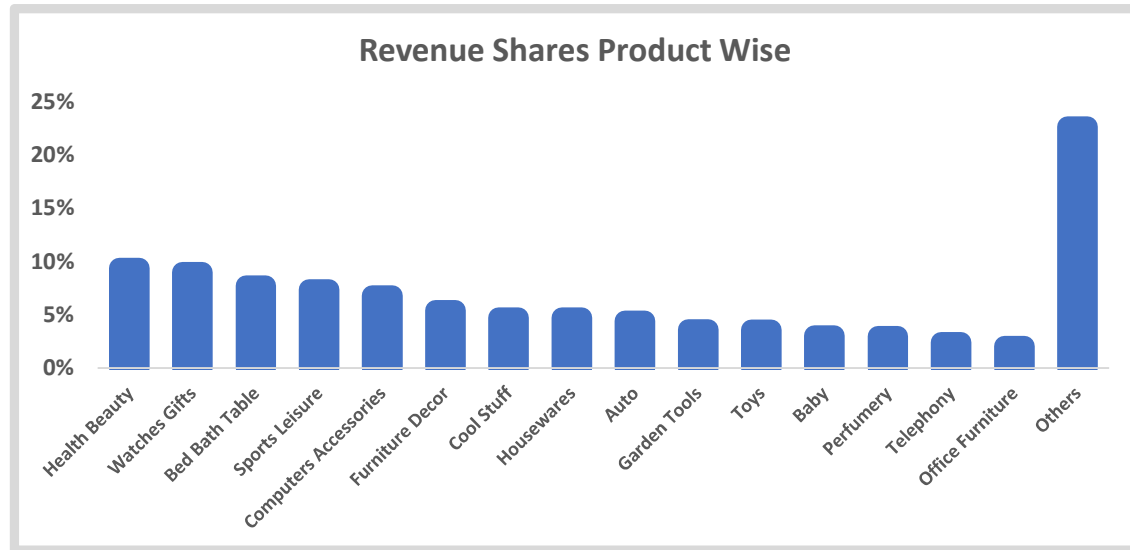
Findings

- The above chart shows that the top 10 most ordered products consists of 64% of total order while remaining categories contribute only 36%.
- Bed Bath Table and Health Beauty are top performing products based on order count.

Business Implication

- These categories are strong customer demand drivers.
 - Marketing budget should focus on maintaining leadership in these categories.
 - Consider cross-selling complementary products to increase average order value.
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Product Category Performance



Findings

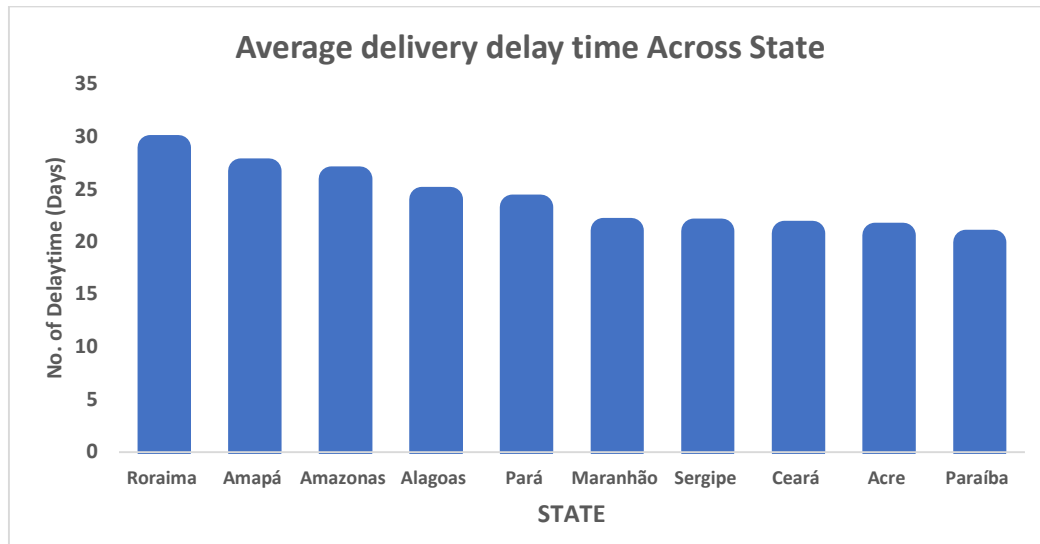
- Revenue is concentrated among a small number of product categories. Out of 71 products only 15 products category makes nearly 77% of revenue.
- Health Beauty and Watches Gifts consists of 18% of total revenue.
- Significant variation exists in category performance.
- Most people purchase bed bath table category product but the revenue generation is mostly from health beauty products.

Business Implications

Inventory planning and promotional strategies should focus on high-performing categories.

7. LOGISTICS PERFORMANCE ANALYSIS

Delivery Performance



Top 10 state with average delivery delay time more than 20 days.

Findings

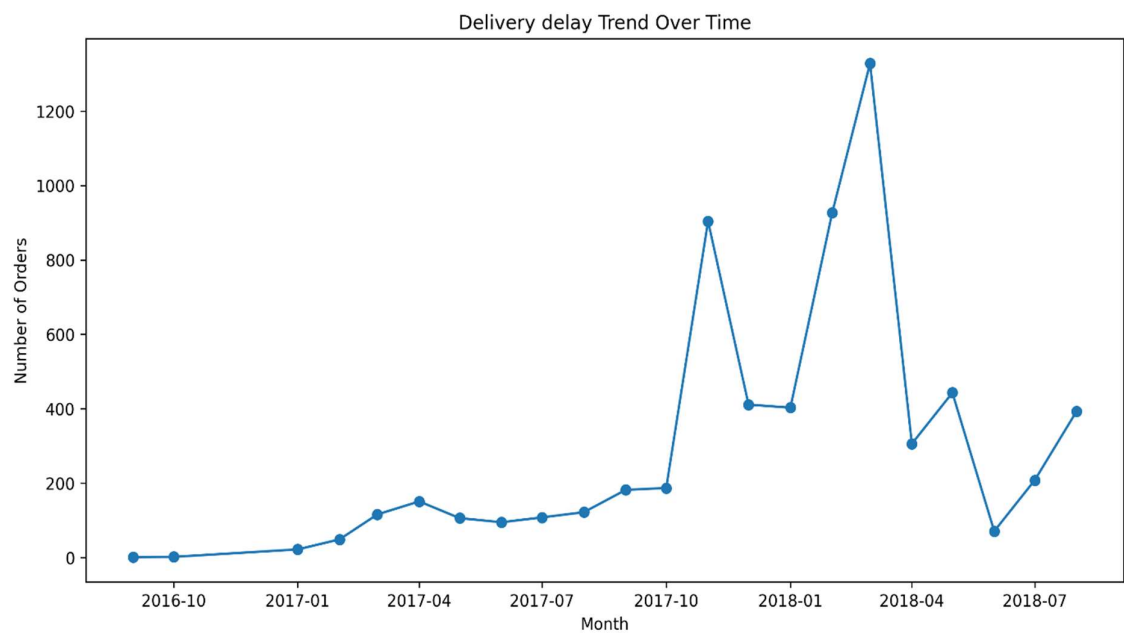
- Delivery duration varies considerably across states. Above region likely suffer from weak logistics infrastructure, longer shipping distances, or limited carrier coverage.
- Delayed deliveries remain a recurring operational issue in some states.

Business Implications

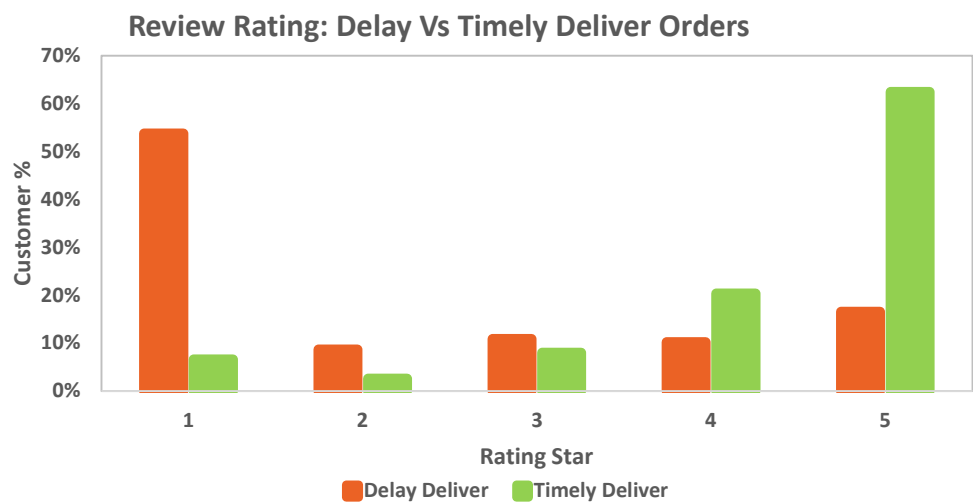
Business should focused on increasing number of delay order over time.

Olist may need regional distribution centers or alternative shipping partners in northern region of Brazil

8. CUSTOMER SATISFACTION ANALYSIS



Review Score Analysis



Findings

- Orders that deliver timely has influence customer positively.
- Delayed deliveries are associated with lower review scores.

Business Implications

Customer satisfaction is strongly linked to operational performance.

Reducing delivery delays can improve customer experience and reduce negative reviews.

9. STATISTICAL VALIDATION

Hypothesis Test 1

Research Question

Do delivery delays affect customer review scores?

Null Hypothesis (H_0)

Delivery delays have no impact on review scores.

Alternative Hypothesis (H_1)

Delivery delays negatively impact review scores.

Statistical Method

Welch's T-Test

Results

Metric	Value
Test Statistic	-100.99
P-Value	0

Conclusion

The null hypothesis is rejected.

The analysis provides statistical evidence that delivery delays significantly reduce customer satisfaction.

Hypothesis Test 2

Research Question

Do high-volume orders generate greater sales?

Null Hypothesis (H₀)

Sales performance is identical across order groups.

Alternative Hypothesis (H₁)

High-volume orders generate significantly greater sales.

Statistical Method

Z-Test

Results

Metric	Value
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Test Statistic	: 3.22
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P-Value	: 0.00125
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Conclusion

The analysis confirms a statistically significant difference between order groups.

10. STRATEGIC RECOMMENDATIONS

Recommendation 1

Improve logistics performance by reducing delivery delays.

Expected Impact:

Higher customer satisfaction and stronger review scores.

Recommendation 2

Increase investment in high-performing regions.

Expected Impact:

Improved revenue generation and customer acquisition.

Recommendation 3

Prioritize inventory planning for top-performing product categories.

Expected Impact:

Higher sales efficiency and improved profitability.

Recommendation 4

Track customer satisfaction KPIs alongside operational KPIs.

Expected Impact:

Earlier identification of customer experience issues.

11. CONCLUSION

This analysis demonstrates that customer satisfaction, operational performance, and revenue generation are closely connected.

Organizations can improve business outcomes through targeted investments in logistics optimization, customer experience management, and category-focused growth strategies.

The findings provide a foundation for future predictive analytics initiatives, including customer segmentation, churn prediction, and sales forecasting.

